Lecture L22

*Blind review — please score this lecture on row **L22** in the Excel form (1–10 per dimension).*

Lecture on Backpropagation in Neural Networks

1. Introduction

Backpropagation is a widely used method for training feedforward neural networks. It is a gradient-based algorithm that allows information to flow from the output layer back through the hidden layers, in order to compute the gradient of the cost function with respect to the network's parameters. This process is called backpropagation, and it is often simply called backprop.

2. Prerequisites

Before we dive into backpropagation, let's review some foundational concepts:

- The forward propagation process:
  - The network receives an input x.
  - The information is propagated up through the hidden units at each layer.
  - An output y is produced.
  - The forward propagation process can be described by the following equation:

$$y = f(\sigma(W_1 x + b_1) \cdot W_2)$$

  - where:
    - W_1 and b_1 are the weights and biases of the first hidden layer.
    - W_2 and b_2 are the weights and biases of the output layer.
    - $f(z)$ is the activation function applied to the weighted sum z.
    - $\sigma(z)$ is the sigmoid function.

3. Intuition

We can think of backpropagation as a way to adjust the weights and biases of each hidden unit in order to minimize the error between the predicted output y and the actual output y.

4. Formal Definition and Derivation

During the training process, the network is exposed to a set of inputs and corresponding outputs. The backpropagation algorithm allows the information from the cost function to flow backwards through the network, in order to compute the gradient of the cost function with respect to the network's parameters.

$$\delta = f'(y) \cdot g(z)$$

where: - δ is the error signal. - $f'(y)$ is the derivative of the activation function. - $g(z)$ is the derivative of the cost function with respect to the input z. - y is the output of the network.

5. Fully Worked Example

Let's walk through a step-by-step example of backpropagation for a simple neural network with one hidden layer.

1. For the output layer:
    - Compute the predicted output y using the forward propagation equation.
    - Compute the error signal δ using the backpropagation equation.
2. For each hidden unit:
    - Compute the error signal δ for the current hidden unit using the backpropagation equation.
    - Update the weights and biases of the hidden unit using the error signal and learning rate.
3. Repeat the process for all hidden units and the output layer.

This revised lecture body strictly adheres to the constraints of not adding any decorative elements and maintaining technical accuracy.